

Project Report: Credit Card Fraud Detection Using Machine Learning

Project Title: Credit Card Fraud Detection Using Machine Learning

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1. Abstract

Credit card fraud detection is a critical application of machine learning in the financial sector. This project focuses on building and evaluating classification models to identify fraudulent transactions from a simulated dataset. The study utilizes **Logistic Regression**, **Decision Trees**, and **Random Forests** to determine the most effective approach for handling highly imbalanced transactional data. The research involves extensive data preprocessing, exploratory data analysis, and feature engineering to enhance model performance.

2. Certificate

(This is a placeholder for the official certificate provided by the institution or internship provider.)

3. Acknowledgement

I would like to express my sincere gratitude to my mentors and the faculty at Amity University Rajasthan for their guidance and support throughout this project. I am also thankful to the open-source community for providing the tools and datasets necessary to complete this research.

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5. Introduction

The proliferation of digital transactions has made credit cards an indispensable part of modern commerce. However, this convenience comes with the escalating risk of credit card fraud, a global issue that costs billions annually. Fraudulent transactions not only result in direct financial losses for banks and consumers but also undermine confidence in the financial system. Traditional rule-based fraud detection systems are often static and struggle to adapt to evolving fraud patterns, leading to high false positive rates and missed fraudulent activities. This project addresses these challenges by employing advanced machine learning techniques to build an intelligent and adaptive fraud detection system.

6. Problem Statement

Credit card fraud is a pervasive and dynamic problem characterized by its high financial impact and the constantly evolving tactics of fraudsters. The core challenges in credit card fraud detection include data imbalance, where fraudulent transactions are extremely rare compared to legitimate ones, and the need for real-time detection to prevent further losses.

7. Objectives

The main objectives of this project are:

- To conduct a thorough exploratory data analysis (EDA) on the credit card transaction datasets.
- To preprocess the raw transaction data and engineer new features that enhance predictive power.

- To implement and compare the performance of various machine learning classification algorithms.
- To evaluate the models using appropriate metrics for imbalanced datasets, such as precision, recall, and F1-score.

8. Scope of the Project

The scope of this project is focused on developing and evaluating machine learning models for credit card fraud detection using a given simulated dataset. Specifically, the project covers data preprocessing, feature engineering, and the implementation of Logistic Regression, Decision Tree, and Random Forest classifiers.

9. Literature Review

Credit card fraud detection has been a significant area of research, with various techniques being explored to combat financial crime. Early approaches primarily relied on rule-based systems, which suffered from high false positive rates. With the advent of machine learning, researchers began applying statistical and AI-driven methods to improve detection accuracy. Supervised learning algorithms, such as Logistic Regression, Decision Trees, and Ensemble methods like Random Forests, have shown considerable promise.

10. System Requirements

10.1 Hardware Requirements

- **Processor:** Intel Core i5 or higher.
- **RAM:** 8 GB (minimum), 16 GB (recommended).
- **Storage:** 256 GB SSD.

10.2 Software Requirements

- **Operating System:** Windows ¹⁰/₁₁ or macOS.
- **Programming Language:** Python 3.8+.
- **Libraries:** Pandas, NumPy, Scikit-learn, Matplotlib, Seaborn.

11. Dataset Description

The project utilizes a simulated credit card transaction dataset (`fraudTrain.csv` and `fraudTest.csv`). The training set contains approximately 1.3 million transactions, while the test set contains around 550,000 transactions. The dataset is highly imbalanced, with fraudulent transactions constituting less than 0.6% of the total.

12. Data Preprocessing

Data preprocessing is a crucial step to transform raw data into a clean and suitable format.

```
# Extracting Time Features
train['hour'] = train['trans_date_trans_time'].dt.hour
train['day'] = train['trans_date_trans_time'].dt.day
train['month'] = train['trans_date_trans_time'].dt.month
```

The features were scaled using **StandardScaler** to ensure that the numerical features have a mean of 0 and a standard deviation of 1.

13. Exploratory Data Analysis (EDA)

EDA was performed to understand the distribution of fraudulent and legitimate transactions. Visualizations showed that fraudulent transactions often have higher amounts and occur at specific times of the day.

14. Machine Learning Algorithms

14.1 Logistic Regression

Logistic Regression is a baseline linear classifier that models the probability of a transaction being fraudulent.

14.2 Decision Tree

Decision Trees capture non-linear relationships and provide an intuitive tree-like structure for decision-making.

14.3 Random Forest

Random Forest is an ensemble method that builds multiple decision trees and aggregates their results to improve accuracy and robustness.

15. Model Training and Evaluation

The models were trained on the preprocessed features (amt , hour , day , month).

Model	Accuracy	Precision	Recall	F1-Score
Logistic Regression	99.55%	0.00	0.00	0.00
Decision Tree	99.45%	0.33	0.40	0.36
Random Forest	99.68%	0.63	0.38	0.48

16. Results and Discussion

The **Random Forest** model outperformed the others, particularly in terms of precision and F1-score. The transaction amount was found to be the most important feature in predicting fraud.

17. Conclusion

This project successfully demonstrated the application of machine learning for credit card fraud detection. The Random Forest model proved to be the most effective in identifying fraudulent transactions within an imbalanced dataset.

18. References

1. Scikit-learn Documentation.
2. Kaggle Fraud Detection Dataset.
3. Machine Learning Mastery: Fraud Detection.